

Continuous Electromagnetic Matter Interrogation Across Conventional Spectral Boundaries

A Federated Frequency-Resolved Architecture with an Extreme-Ultraviolet-to-Soft-X-Ray Demonstration

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ABSTRACT

The electromagnetic spectrum is physically continuous, while the instruments used to generate, select and detect radiation are necessarily specialized by spectral regime. This paper proposes Continuous Electromagnetic Matter Interrogation (CEMI), implemented conceptually as a federated Electromagnetic Tuning Array (ETA): neighboring measurement modules retain the source, optics and detector technology appropriate to their own energy intervals, but are joined through deliberate spectral overlap, cross-calibration, quantified uncertainty and a common photon-energy-indexed data model. The central hypothesis is that this architecture can improve the comparability of material-response measurements across conventional spectral boundaries and may reveal reproducible spectral structure that is less evident when datasets are acquired and interpreted as isolated instrument-specific segments. “Continuous” therefore denotes densely sampled, overlapping and cross-calibrated experimental coverage within stated resolution and uncertainty; it does not require one uninterrupted tunable source. A first demonstration is proposed over approximately 70-200 eV, using well-characterized aluminium and silicon L-shell absorption-edge regions as calibration landmarks. The paper does not predict a new electromagnetic band or assume that novel physics must occur at a named band boundary. Its contribution is a falsifiable, boundary-first measurement architecture intended to separate genuine sample response from source, detector and calibration discontinuities.

Keywords: electromagnetic spectrum; EUV; XUV; soft X-ray; spectroscopy; cross-calibration; absorption edges; frequency-resolved measurement; measurement continuity

Core hypothesis

If the same material system is interrogated across a conventional spectral boundary using densely sampled, overlapping measurements that are cross-calibrated and represented on a common energy axis, reproducible frequency-dependent features may become easier to distinguish from instrument-specific discontinuities and may reveal structure that is less evident in independently acquired spectral segments.

What this paper does NOT claim

It does not propose an undiscovered band between ultraviolet and X-rays, a universal radio-to-gamma source, or a new law of radiation-matter interaction. The spectrum is continuous; the research question concerns continuity of measurement, calibration and interpretation.

1. Introduction

Mechanical tuning forks provide an intuitive example of frequency-selective behaviour: geometry and material properties create a strong mechanical resonance around a characteristic acoustic frequency. Electromagnetic radiation cannot be treated with a literal mechanical analogue across all wavelengths, but the underlying intuition - interrogating physical systems at controlled frequencies - maps naturally onto spectroscopy, resonant cavities, optical frequency combs, tunable sources and energy-selective detectors.

The electromagnetic spectrum itself is continuous. The commonly named regions - radio, microwave, infrared, visible, ultraviolet, X-ray and gamma-ray - are operational classifications rather than physical breaks in the spectrum [1,2]. Frequency, wavelength and photon energy are linked by the standard relations:

$$E = h\nu = hc/\lambda$$

E = photon energy; ν = frequency; λ = wavelength; h = Planck constant; c = speed of light.

The instruments used to work with different parts of the spectrum are not continuous in the same sense. Source physics, optical components, detector materials, vacuum requirements, calibration methods and uncertainty models change substantially with energy. CEMI asks whether these necessary hardware transitions can be organized so that the resulting measurements remain physically comparable across a shared energy coordinate.

2. Scientific context

2.1 Frequency-resolved electromagnetic measurement is established

Optical frequency combs provide a particularly useful conceptual analogue. They create highly regular frequency markers and enable precise optical-frequency measurement. Diddams, Vahala and Udem describe frequency-comb technology as a means of coherently connecting radio-frequency references with wide optical bandwidths [3,4]. JILA/NIST work has also demonstrated frequency-comb spectroscopy in the extreme-ultraviolet (EUV) [5]. CEMI therefore does not claim invention of frequency-resolved electromagnetic metrology.

2.2 EUV and soft-X-ray spectroscopy already probe rich material response

Near-edge X-ray absorption methods deliberately vary photon energy through core-level thresholds because the energy-dependent response contains information about chemical state, electronic structure and local environment. NIST describes NEXAFS measurements using a tunable focused soft-X-ray beam over roughly 0.1-2.2 keV, with electron and fluorescence detection modes [6]. XAFS similarly measures absorption as incident energy is tuned relative to elemental binding energies [7].

Modern high-harmonic-generation and free-electron-laser sources extend coherent ultrafast methods into EUV and X-ray regimes, enabling element- and orbital-selective spectroscopy [8]. Ultrafast EUV-to-soft-X-ray measurements are also increasingly used in liquids and condensed matter [9,11]. The proposal here therefore builds on mature spectroscopic physics rather than replacing it.

3. Operational definition of “continuous”

Definition used throughout this paper

In this paper, “continuous” does not necessarily mean interrogation by a single uninterrupted tunable source. It refers to measurement continuity in the resulting physical coordinate: densely sampled photon-energy measurements obtained from one or more overlapping spectral modules, cross-calibrated within shared energy intervals and combined with quantified uncertainty into a common energy-indexed dataset. Continuous therefore means continuous experimental coverage and comparability within stated resolution and uncertainty, not

continuous hardware.

For sampled energies:

$$E_1 < E_2 < \dots < E_n$$

Measurements may originate from modules $M_1, M_2, \dots, M_k$. Neighboring modules should have a non-empty shared calibration interval:

$$\Omega_{i,i+1} = E(M_i) \cap E(M_{i+1}) \neq \emptyset$$

This definition deliberately separates three concepts that are often conflated: (i) physical continuity of the electromagnetic spectrum, (ii) sampling density and instrumental resolution within a dataset, and (iii) continuity of hardware or source mechanism. CEMI requires the first two to be represented coherently; it does not require the third.

Measurement continuity through deliberate overlap

“Continuous” refers to coverage and comparability - not one universal source.

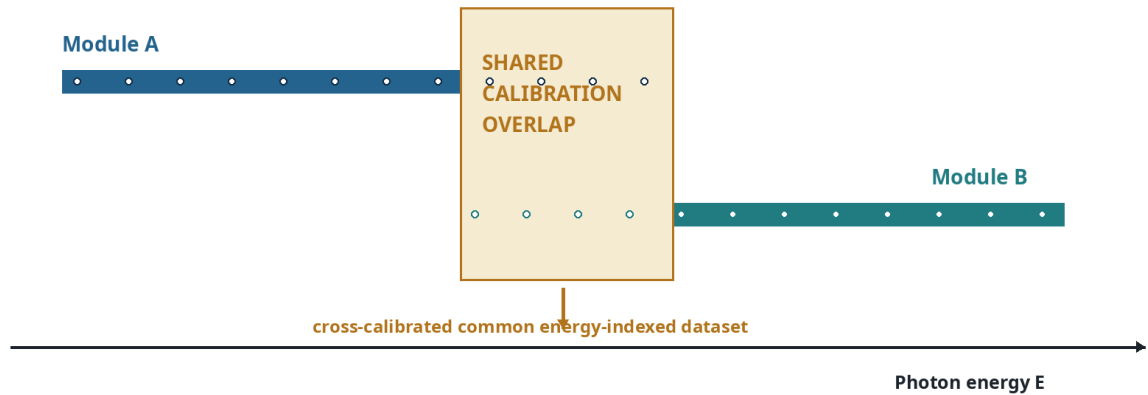


Figure 1. “Continuous” CEMI coverage is produced by deliberate overlap and cross-calibration between adjacent modules, not by requiring one universal source.

4. Research gap and proposed contribution

The research gap is not a missing set of electromagnetic frequencies. Instead, the proposed gap is experimental fragmentation: neighboring spectral regimes can be investigated with different sources, selectors, detectors, calibration standards, sampling grids, uncertainty conventions and data structures. When such datasets are combined only after acquisition - or not combined at all - a physically smooth response may appear discontinuous, while a real localized feature may be difficult to separate from an instrumental transition.

Table 1. Established capabilities versus the proposed CEMI emphasis

Area	Established capability	CEMI / ETA emphasis
Frequency selection	Tunable lasers, monochromators, frequency combs	Coordinate neighboring systems on a common energy axis
Spectroscopy	NEXAFS/XAFS, absorption, emission, scattering	Make conventional instrument boundaries deliberate overlap zones

Calibration	Instrument-specific standards and references	Require shared reference measurements in overlap intervals
Data representation	Facility- or technique-specific formats	Common minimum measurement record with provenance and uncertainty
Anomaly handling	Technique-specific interpretation	Evidence ladder separating noise, artifacts, known physics and unexplained effects

4.1 Potentially distinctive contribution

The potentially distinctive contribution is the combination of five design principles: boundary-first experiment design, federated instrumentation, explicit overlap, common response semantics, and uncertainty-aware cross-module stitching. None of these principles should be presented as individually unprecedented. The research question is whether their systematic integration produces measurably better continuity and interpretability at boundaries between conventional instrument classes.

5. Research questions and falsifiable hypotheses

5.1 Primary research question

RQ1

What measurable changes occur in a defined material system when photon energy is sampled across a conventional spectral transition using overlapping measurements, consistent calibration and a common response model, rather than treating the instrument boundary as an endpoint?

5.2 H1 - measurement-continuity hypothesis

For a defined material system, densely sampled photon-energy interrogation across a conventional spectral boundary, normalized through overlapping cross-calibration, can produce a more internally consistent response curve and may reveal reproducible spectral structure that is less distinguishable when the same energy interval is measured and interpreted as independent segments.

5.3 H2 - calibration hypothesis

Shared reference measurements within overlap intervals can reduce or quantify cross-instrument normalization error sufficiently to distinguish some genuine sample-response changes from detector, source or calibration discontinuities. This is particularly important near absorption edges, where uncertainty may increase and model sensitivity can be high [13].

5.4 H3 - multimodal-correlation hypothesis

A feature observed in two or more independently measured response channels at the same photon energy is more likely to represent a sample-related effect than a feature confined to a single detector channel, provided systematic correlations are controlled.

$$R(E) = [T(E), A(E), F(E), P(E), S(E)]$$

Example response vector: transmission, absorption, fluorescence, photoelectron response and scattering. Not all channels are required in every experiment.

6. Federated architecture: the Electromagnetic Tuning Array

ETA is an architectural name for the CEMI implementation concept. The “tuning” analogy refers to selecting and comparing defined electromagnetic energies, not to a literal fork and not to one resonator mechanism spanning the spectrum. Each module remains optimized for its spectral interval; the federation occurs at calibration, metadata and analysis layers.

Federated CEMI / Electromagnetic Tuning Array architecture

Specialized hardware, shared physical coordinate, explicit overlap and calibration

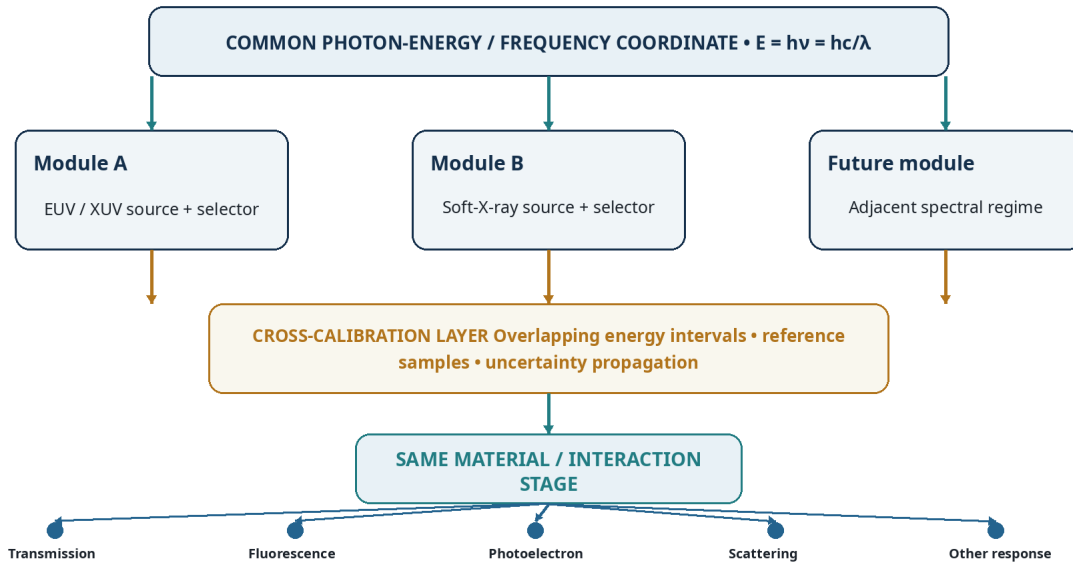


Figure 2. Federated architecture. Specialized modules retain appropriate source and detector physics while sharing energy coordinates, cross-calibration and response semantics.

6.1 Minimum measurement record

A practical common data model should record enough context to compare measurements acquired by different modules. One minimal conceptual record is:

$$M = \{E, \nu, \lambda, \Delta E, \Phi, P, t, G, S, R, \sigma, C\}$$

Symbol	Meaning	Symbol	Meaning
E	photon energy	ν	frequency
λ	wavelength	ΔE	instrumental energy resolution
Φ	incident photon flux	P	polarization state
t	pulse duration or integration interval	G	geometry and detector configuration
S	sample state and conditions	R	measured response
σ	measurement uncertainty	C	calibration and provenance metadata

7. Cross-module calibration and stitching

Adjacent modules should be designed with a shared energy interval that is large enough to contain multiple calibration points and, where feasible, reference features. Both modules measure the same reference sample under documented conditions. The goal is not to force identical raw signal levels, but to estimate the mapping from each instrument's response to a common normalized representation.

An illustrative normalized response is:

$$R_i(E) = [S_i(E) - B_i(E)] / [\Phi_i(E) \cdot G_i(E)]$$

where S is measured signal, B is background, Φ is incident photon flux and G represents detector/geometry corrections. A simple cross-module consistency statistic can be written:

$$Z(E) = |R_A(E) - R_B(E)| / \sqrt{\sigma_A^2(E) + \sigma_B^2(E)}$$

This expression is illustrative rather than prescriptive. A real experiment would use a calibration model appropriate to its detector statistics, correlated systematics and spectral resolution. The important requirement is that disagreement inside the overlap interval be quantified before the datasets are stitched.

8. First demonstration: approximately 70-200 eV

The first experiment should be intentionally narrow. A practical starting window is approximately 70-200 eV, with exact limits set by source coverage, photon flux, optics, detector response, instrumental resolution and required overlap. The range contains well-characterized aluminium and silicon L-shell absorption-edge regions that can function as physical landmarks for calibration and validation [12]. It also spans energy intervals for which material-dependent fine structure can be measured with short-wavelength spectroscopy.

Important qualification on edge energies

The NIST X-Ray Transition Energies Database identifies the quoted aluminium and silicon values as L₂ and L₃ absorption edges. They should be treated as reference edge regions, not universal fixed positions of every solid-state spectral feature. Measured spectra may also depend on chemical state, bonding, sample structure and measurement method.

Table 2. Selected NIST experimental L-edge values used as reference landmarks [12]

Element / edge	Direct experimental (eV)	Combined experimental (eV)	Use in CEMI demonstration
Al L ₂	72.760(63)	72.75(20)	low-energy validation landmark
Al L ₃	72.760(63)	72.870(61)	low-energy validation landmark
Si L ₂	100.8(12)	99.20(10)	cross-check near ~100 eV
Si L ₃	100.8(12)	99.340(61)	cross-check near ~100 eV

8.1 Candidate samples

A conservative validation sequence should use familiar materials before exotic samples:

- aluminium metal
- aluminium oxide
- silicon

- silicon dioxide
- reference substrate / blank

The purpose of the first demonstration is to test the architecture, calibration and stitching logic. A discovery-oriented sample campaign should follow only after the system reproduces accepted edge structure and uncertainty behaviour.

8.2 Adaptive sampling

Sampling should be dense enough to resolve the expected instrumental line shape and known features, but numerical step size must not be confused with physical resolution. When a preliminary scan indicates a feature, the system may increase sampling density locally, repeat the scan, acquire an additional response channel and perform a local calibration check.

9. Experimental protocol

Boundary-first experimental workflow

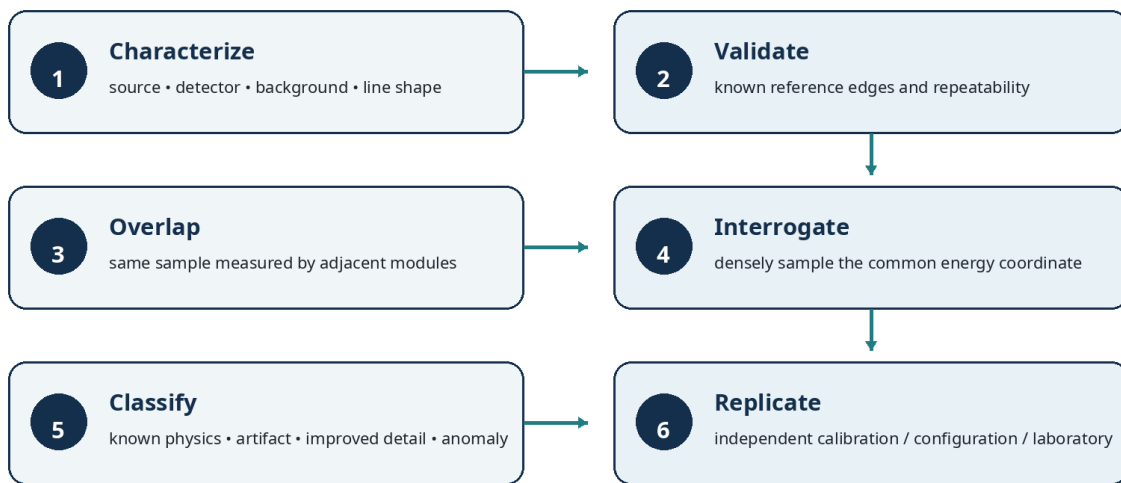


Figure 3. Six-stage boundary-first protocol from instrument characterization to independent replication.

Phase 1 - Instrument characterization

Measure source spectrum, incident flux stability, detector response, background, repeatability, energy calibration and instrumental line shape.

Phase 2 - Reference validation

Measure known spectral landmarks and compare position, shape and uncertainty with accepted references. Confirm that any disagreement is understood before proceeding.

Phase 3 - Cross-module overlap

Measure the same sample in the shared interval using both neighboring modules. Estimate normalization and uncertainty mappings.

Phase 4 - Common-axis interrogation

Construct the merged response on one photon-energy coordinate while retaining source-module identity and provenance for every data point.

Phase 5 - Candidate-feature analysis

Test unexpected peaks, shoulders, threshold behaviour, correlated channel changes or model residuals against instrumental and sample-state explanations.

Phase 6 - Independent verification

Repeat with altered scan order, independent calibration, blanks, additional samples and - for important candidates - a second instrument or laboratory.

10. Feature classification and evidence standard

A boundary-focused experiment creates a risk of over-interpreting artifacts. CEMI therefore treats anomaly classification as part of the experimental design rather than an afterthought.

Evidence ladder for unexpected spectral features

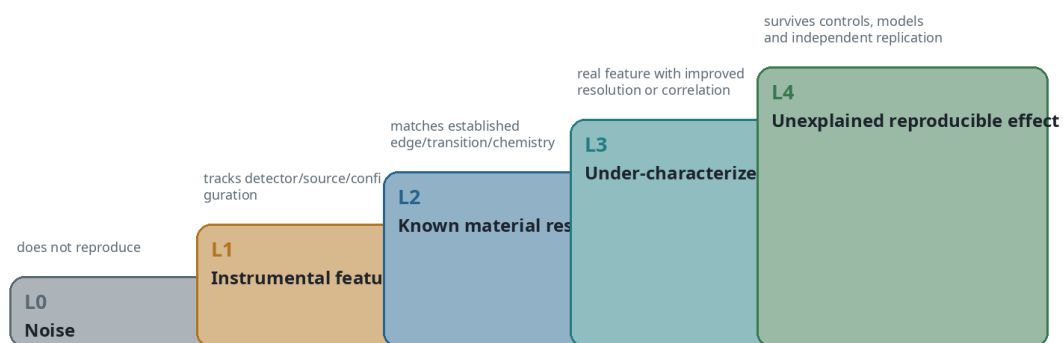


Figure 4. Evidence ladder. Only effects that survive controls and independent replication progress to the strongest category.

L0 - Noise

The feature is not repeatable within the stated uncertainty.

L1 - Instrumental feature

The feature follows a source, detector, calibration or configuration change.

L2 - Known material response

The feature matches established absorption-edge, transition, chemical-state or structural behaviour.

L3 - Under-characterized response

The feature is physical and reproducible but primarily represents improved resolution, calibration or cross-channel correlation of known mechanisms.

L4 - Unexplained reproducible effect

The feature survives instrument swaps or independent calibration, sample controls and comparison with accepted models. Only this category warrants investigation as potentially new physical behaviour.

11. Falsifiability and decision criteria

The concept is useful only if it can fail. H1 and the broader architecture would be weakened if controlled testing shows that segmentation across instrument regimes adds no meaningful uncertainty or interpretive ambiguity and that federation provides no measurable information gain.

- Cross-calibrated stitching does not improve agreement in overlap regions beyond standard independent workflows.
- Candidate boundary features disappear under repeat measurement, reversed scan order, altered configuration or improved calibration.
- All apparent discontinuities are adequately explained by known detector, source, sample-damage or normalization effects.
- The merged representation introduces greater uncertainty than separate best-in-class analyses.
- The operational cost and complexity of overlap and federation exceed any reproducible improvement in calibration or scientific inference.

A negative result would still be informative: it would quantify how effectively current segmented spectroscopy already preserves cross-regime information.

12. Expected outcome classes

Outcome	Interpretation
A. Architecture validated; no additional physical features	The system improves traceability or data interoperability but recovers only established spectroscopy.
B. Improved characterization	Known transitions or chemical-state effects become more internally consistent or better correlated across channels.
C. Instrumental discontinuity identified	An apparent material feature is shown to arise from source, detector or calibration boundaries.
D. Previously unresolved material-response detail	A reproducible feature survives controls but is best interpreted as improved characterization within known physics.
E. Unexplained reproducible effect	A feature survives cross-instrument checks, sample controls and accepted models, motivating targeted theory and independent replication.

13. Limitations and experimental cautions

- No single source, optical train or detector can provide equivalent performance across the entire electromagnetic spectrum; federation is a design necessity, not a temporary inconvenience.
- Short-wavelength irradiation may alter or damage samples. Dose, scan order and sample history must therefore be recorded as part of the measurement state.
- Photon flux, detector efficiency and signal-to-noise can change sharply across neighboring regimes, complicating normalization.

- Dense sampling does not create resolution beyond the instrumental line shape. The dataset must report both step size and true resolving power.
- Near-edge tabulations and attenuation models carry non-negligible uncertainty. NIST uncertainty guidance is explicitly energy- and region-dependent [13].
- Surface contamination, oxidation, temperature, polarization, geometry, thickness and chemical environment may shift or reshape measured features.
- A common data schema cannot eliminate technique-specific physics; it must preserve module-specific metadata rather than forcing false equivalence.

14. Extension beyond the first demonstration

If the EUV/soft-X-ray demonstration shows measurable benefit, the same architecture could be tested at other neighboring instrumentation regimes. This should proceed boundary by boundary rather than by claiming immediate radio-to-gamma continuity. Each extension would define its own source technologies, detector physics, calibration standards, overlap interval and expected interaction mechanisms.

Candidate boundary	Primary research task
Radio / microwave	Define overlap, calibration transfer, uncertainty model and material-response channels.
Microwave / terahertz	Define overlap, calibration transfer, uncertainty model and material-response channels.
Terahertz / infrared	Define overlap, calibration transfer, uncertainty model and material-response channels.
Infrared / visible	Define overlap, calibration transfer, uncertainty model and material-response channels.
Visible / ultraviolet	Define overlap, calibration transfer, uncertainty model and material-response channels.
EUV / soft X-ray	Define overlap, calibration transfer, uncertainty model and material-response channels.
Soft / hard X-ray	Define overlap, calibration transfer, uncertainty model and material-response channels.
Hard X-ray / gamma	Define overlap, calibration transfer, uncertainty model and material-response channels.

15. Scientific and engineering significance

The strongest value proposition is methodological. The spectrum itself does not need a new category; rather, measurement systems may benefit from being designed around physical continuity instead of administrative or historical band labels. A successful CEMI implementation could provide:

- a reproducible way to transfer calibration across neighboring spectral instruments;
- a common representation for response, resolution, uncertainty and provenance;
- a clearer distinction between genuine material features and instrument-boundary artifacts;
- a framework for multimodal correlation at the same photon energy; and
- a disciplined path from exploratory anomaly detection to independent physical verification.

16. Conclusion

The electromagnetic spectrum is physically continuous, while experimental access to it is technologically segmented. Continuous Electromagnetic Matter Interrogation reframes that segmentation as an explicit measurement problem. It proposes that neighboring spectroscopy modules be designed to overlap, measure common references, propagate uncertainty and contribute to a shared photon-energy-indexed dataset.

The first demonstration is deliberately modest: approximately 70-200 eV, with aluminium and silicon L-shell absorption-edge regions used as known landmarks rather than discovery claims. The architecture succeeds only if cross-calibration and common-axis analysis provide measurable benefit over independent best-practice measurements.

Central proposition

The electromagnetic spectrum may contain no missing frequencies while experimental practice can still contain missing continuity - in sampling, calibration, uncertainty transfer and cross-regime correlation. Restoring that continuity is a testable engineering and scientific hypothesis.

If the architecture reveals only known physics more cleanly, that is still a valid outcome. If an unexpected feature survives calibration, control experiments and independent replication, only then should it be escalated as a candidate for new or under-characterized physical behaviour.

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Reference integrity note. Reference 13 uses the exact NIST page title. Although that webpage was created and updated in 2026, the parent NIST Standard Reference Database 66 reports that its scientific data content was last updated in January 2005; the paper therefore does not describe the underlying uncertainty data as “updated 2026.”